ECE 310 Fall 2026
Lecture 18: Ideal Digital-to-Analog Conversion

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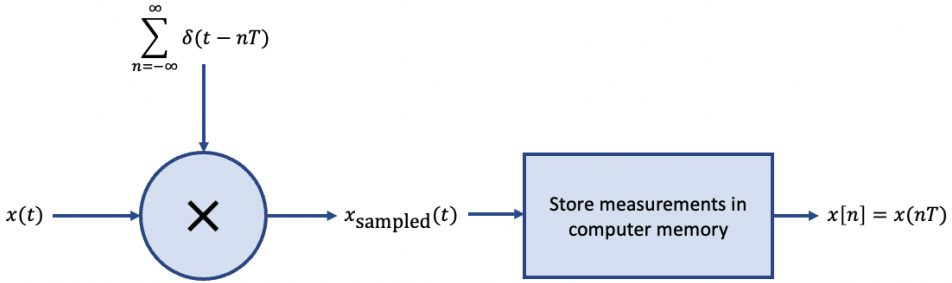
Learning Objectives

After this lecture, you should be able to:

- Explain ideal digital-to-analog conversion using sinc interpolation.
- Describe ideal digital-to-analog conversion in the frequency domain via ideal low-pass filtering.



Recap of Ideal Analog-to-Digital Conversion



$$X_s(\Omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_a\left(\Omega - \frac{2\pi}{T}k\right)$$

$$X_d(\omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_a\left(\frac{\omega - 2\pi k}{T}\right)$$

$$\omega = \Omega T$$

How can we reconstruct $x_a(t)$ from $x[n]$?

For a given sampling period T, we may re-associate samples from $x[n]$ in time:

$$x[n] = x_a(nT).$$

What is $x_a(t)$ when $t \neq nT$?



Interpolation

The process of "filling in the blanks" or "connecting the dots" is known as *interpolation*. The interpolation of digital samples to recover an analog signal can be expressed using the following reconstruction formula.

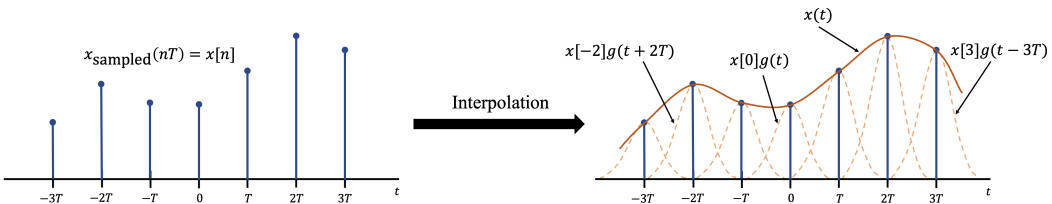
$$x_a(t) = \sum_{n=-\infty}^{\infty} x[n]g(t - nT)$$



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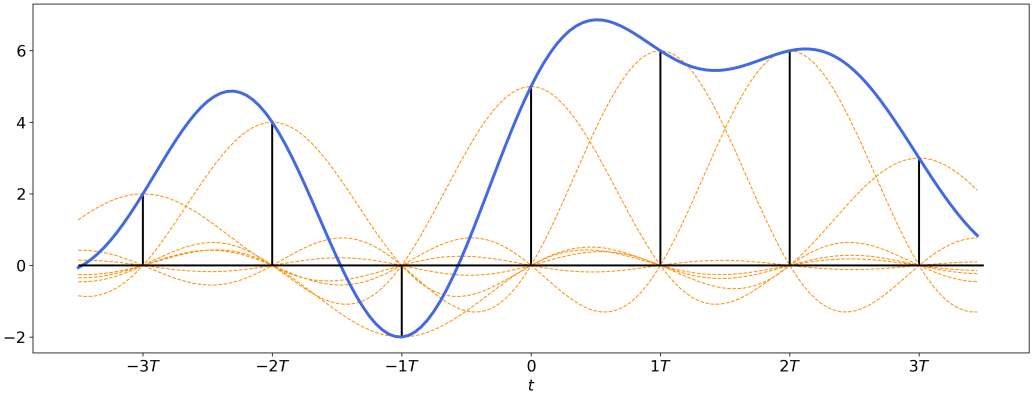
Ideal Digital-to-Analog Conversion

What is the correct choice for $g(t)$ to perfectly recover bandlimited analog signals?



Ideal Sinc Interpolation: Time-domain

$$x(t) = \sum_{n=-\infty}^{\infty} x[n]\text{sinc}\left(\frac{\pi}{T}(t - nT)\right)$$

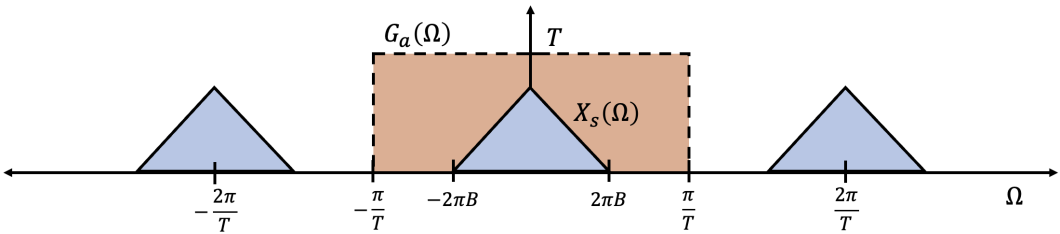


Ideal Sinc Interpolation: Frequency-domain

$$X_a(\Omega) = X_s(\Omega) G_a(\Omega)$$

$$X_s(\Omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_a\left(\Omega - \frac{2\pi}{T} k\right)$$

$$G_a(\Omega) = T \operatorname{rect}\left(\frac{\Omega}{\frac{2\pi}{T}}\right)$$



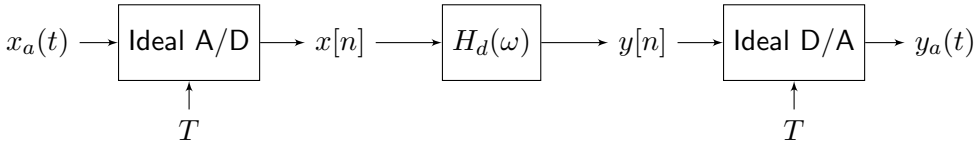
Example: Sketching Ideal D/A Conversion

Let $x[n] = 3\delta[n-1] + \delta[n-4]$. Sketch $x_a(t)$ if we recover $x[n]$ by ideal sinc interpolation with sampling period $T = 2$ seconds.

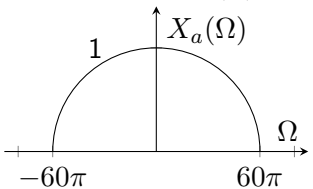


Overall Sampling System Example

Consider the below sampling system:



We have that $H_d(\omega)$ is a low-pass filter with cutoff $\omega_c = \frac{\pi}{2}$ and $T = \frac{1}{80}$s.



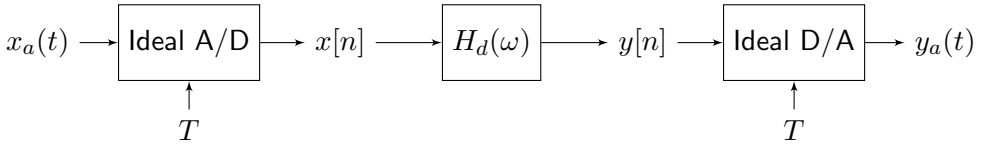
(a) Sketch $X_d(\omega)$.

(b) Sketch $Y_d(\omega)$.

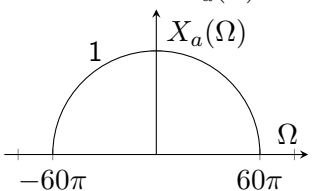
(c) Sketch $Y_a(\omega)$.

Overall Sampling System Example

Consider the below sampling system:



We have that $H_d(\omega)$ is a low-pass filter with cutoff $\omega_c = \frac{\pi}{2}$ and $T = \frac{1}{50}$s.



(a) Sketch $X_d(\omega)$.

(b) Sketch $Y_d(\omega)$.

(c) Sketch $Y_a(\omega)$.